

REQUEST FOR INFORMATION

AMENDMENT 1

Missile Defense System - Next (MDS-Next) Capability Solutions for Strategic Architecture Thrust Areas

Notice Number: 26-MDA-12338

NAICS Code: 541715 - Research and development in the physical, engineering, and life sciences, except nanotechnology and biotechnology

1.0 BACKGROUND

The Missile Defense Agency (MDA) is executing a strategic planning effort to inform future capability development and investment opportunities in response to the 2035+ projected threat environment. As part of this effort, the Agency has identified Strategic Architecture Thrust Areas representing priority focus areas for advancing missile defense capabilities in the 2035+ timeframe. These thrust areas reflect where the Agency is seeking new or alternative approaches to address future mission challenges and achieve improved mission outcomes.

This RFI represents the transition from identifying and prioritizing strategic focus areas to evaluating solution approaches that can advance those areas. Responses will inform prioritization of capability areas and potential POM investment decisions.

Detailed descriptions of the Strategic Architecture Thrust Areas will be available to respondents with the appropriate clearance level. Additional context on future capability attributes is provided in the MDS-Next Characteristics briefing that will be available to respondents at the Controlled Unclassified Information (CUI) level. See Section 9.0 for instructions on requesting access to these resources.

2.0 OBJECTIVE

The objective of this RFI is to obtain solution-oriented information describing specific capability approaches and architectural concepts that advance one or more of the identified Strategic Architecture Thrust Areas. Responses should clearly describe what capability is being proposed, how it addresses the thrust area, and how it improves mission outcomes.

Information received will be used to:

- Identify solution approaches that measurably address validated Strategic Architecture Thrust Areas
- Assess the technical feasibility, maturity, and integration considerations of proposed approaches
- Enable focused technical dialogue and evaluation at the 1:1 technical exchange
- Inform prioritization of potential capability areas and support POM-28 investment decisions

This effort is intended to produce decision-relevant insights to further long-range planning efforts.

3.0 STRATEGIC FRAMING

The Agency is not solely seeking incremental improvements to existing systems or approaches. Respondents are encouraged to consider whether fundamentally different capability constructs, architectures, or operational concepts are required to advance the identified Strategic Architecture Thrust Areas.

In particular, respondents should consider:

- Whether current approaches represent the most effective and efficient path forward, or if alternative paradigms may provide greater mission advantage
- What new capabilities or concepts, including those developed outside traditional missile defense applications, could be adapted to significantly improve outcomes
- Preference will be given emerging technologies, cross-domain solutions, or non-traditional approaches that could change how the mission is executed or approached

The intent is to identify both evolutionary improvements and non-traditional or disruptive approaches that provide meaningful operational advantage.

4.0 SCOPE

Responses should focus on capability-level approaches and architectural concepts that:

- Directly align to one or more Strategic Architecture Thrust Areas
- Address challenges posed by the 2035+ projected threat environment
- Demonstrate the potential to improve operational effectiveness at the system or mission level

Respondents should consider:

- How the proposed approach changes outcomes relative to the identified Strategic Architecture Thrust(s), and if applicable, the Key Area(s) where the architecture would particularly benefit from additional science and technology (S&T)
- Applicability across multiple mission areas or threat sets, where relevant
- Integration with existing and planned missile defense architectures

5.0 SUBMISSION REQUIREMENTS

Respondents shall use the Strategic Architecture Thrust Areas to ensure clear alignment of proposed solutions. The MDS-Next Characteristics briefing is provided as additional context to inform how proposed approaches align to desired future mission attributes. The characteristics briefing is not an exhaustive list and is provided as reference. Characteristics provided in the briefing should not be interpreted as limiting or constraining potential solutions.

Submissions must clearly align to one or more Strategic Architecture Thrust Areas. Responses should be specific, structured, and focused on capability outcomes. Respondents should avoid general descriptions, marketing material, or platform-centric responses. Responses that do not demonstrate clear alignment or sufficient detail may be assessed as less relevant.

Responses shall be submitted as concise pitch papers and must address the following elements:

- **Strategic Architecture Thrust Area Alignment**

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- Identify the specific Strategic Architecture Thrust(s) and Key Area(s) addressed and describe how the proposed approach advances that intent
- Maintain clear traceability to the Strategic Architecture Thrust(s) and Key Area(s) throughout the response
- **Capability Approach**
 - Provide a clear description of the proposed capability or architectural concept
 - Describe:
 - The function of the capability
 - How it operates within the broader system
 - Responses should address:
 - Whether the approach represents a new or alternative way to achieve the mission outcome
 - How it differs from or improves upon current or planned capabilities
 - Estimated timeline to achieve initial operational fielding of the proposed capability, including key milestones for development, prototyping, testing, and deployment
 - Where applicable, identify:
 - Approaches derived from other domains, missions, or industries
 - Capabilities not traditionally associated with missile defense that could be adapted to this mission
- **Operational Impact**
 - Describe how the approach improves mission outcomes in relation to the thrust area
 - Address:
 - What operational limitations are mitigated or eliminated
 - Whether the approach enables new operational concepts or reduces reliance on existing constructs
 - Clearly articulate whether the approach:
 - Enhances current capabilities, or
 - Fundamentally changes how the mission is performed
- **Technical Feasibility and Maturity**
 - Describe the current level of maturity (concept, prototype, demonstration, etc.)
 - Technology Readiness Level
 - Identify key technical challenges, risks, or dependencies
- **Cost Considerations**
 - Provide the estimated Rough Order of Magnitude (ROM) cost for the proposed solution, including development, prototyping, integration, production, and fielding, as applicable
 - Identify:
 - Major cost drivers and underlying assumptions
 - Any dependencies on Government-furnished resources, infrastructure, or enabling capabilities
 - Areas of cost uncertainty or variables that could materially affect the estimate

- **Integration Considerations**
 - Describe how the approach would integrate with:
 - Existing systems
 - Planned architectures
 - Identify required interfaces, dependencies, and enabling capabilities
- **Small Business Considerations**
 - The Government is seeking to understand the capability of small businesses to contribute to the Strategic Architecture Thrust Areas and to inform potential future acquisition strategies.
 - Small business respondents are requested to address the following:
 - Describe relevant capabilities, technologies, or solution approaches the small business can provide that align to the Strategic Architecture Thrust Areas
 - Identify any unique or innovative contributions the small business brings, including capabilities derived from non-traditional or commercial applications
 - If proposing a teaming arrangement, describe:
 - The role of the small business in the overall solution
 - Any teaming partners or subcontractors, as applicable
 - How the small business contributes to the overall capability approach
- **Additional Considerations**
 - Respondents are strongly encouraged to identify solutions that:
 - Leverage capabilities from other Services, Agencies, or Departments
 - Adapt technologies or concepts from the commercial sector or other mission areas
 - Reduce costs per missile defense kill or sustainment costs of fielded system
 - Represent cross-domain or non-traditional applications relevant to missile defense

6.0 USE OF SUBMISSIONS

Responses will be reviewed by the Agency to identify Technology Areas for further examination and will inform prioritization of capability areas and potential POM investment decisions. The submissions will be discussed as part of the respondent's requested 1:1 technical exchange. Submission does not constitute a commitment by the Agency to pursue any specific approach or activity.

Provided responses will not be returned and respondents are advised that MDA is under no obligation to acknowledge receipt of the information received except as required to confirm receipt of classified materials, or to request clarification with respect to any information that is submitted in accordance with the instructions contained in this RFI. MDA makes no commitment regarding the use of a submission; nor is any acquisition expected from this RFI or the submissions that may be made hereunder.

7.0 MARKING

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Respondents are encouraged to respond with the submission of information that is not constrained by proprietary information rights. If proprietary information is included in a respondent's submission, the respondent is responsible for ensuring its submission contains appropriate and adequate markings of said proprietary information. Further, any proprietary information **MUST** be marked as such on a page-by-page basis.

The Government intends to use third party, non-Government (i.e. contractor, Federally Funded Research, and Development Center personnel) support personnel as subject matter experts in the review of responses received, including the review of any marked or unmarked proprietary information provided. MDA utilizes subject matter experts at Federally Funded Research Laboratories and other support contractors to meet our mission objectives. The Agency maintains non-disclosure agreements on file with these organizations and companies to allow them access to proprietary information for the purpose of evaluations and recommendations. Organizational Conflict of Interest (OCI) and Personal Conflict of Interest (PCI) clauses are incorporated within each of these contracts. Submission of a response to this RFI constitutes the participant's acknowledgement and agreement that the information provided in the response, including any marked or unmarked proprietary or source selection information, may be disclosed to these non-Government support personnel. Respondents with questions or objections could contact the following unclassified email mda.mbx.sppa@mail.mil.

Respondents are further notified the Government may use Artificial Intelligence (AI) mechanisms to review and assess responses.

8.0 COMPENSATION

This RFI only supports MDA planning activities. Submission of information by a respondent hereunder is entirely voluntary. The Government does not intend to award a contract based on this request or reimburse any costs for the information requested as a result of any company, organization, or individual providing a response to the RFI. MDA reserves the right to use information that has been voluntarily provided by respondents without compensation as part of MDA's ongoing efforts to refine the Missile Defense System Strategic Investment Plan and Technology Roadmap.

9.0 RFI RESOURCES

The following resources are available to assist respondents in replying to this RFI:

- Strategic Architecture Thrust Areas briefing (SECRET level)
- MDS-Next Characteristics briefing (CUI level)
- 1:1 technical exchange with subject matter experts from MDA

To request access to these resources, send an unclassified email to mda.mbx.sppa@mail.mil with the information in a. – d. listed below. Complete requests must be received by MDA no later than (NLT) 1200 Central Standard Time (CST) on 1 May 2026.

a. General Information

- Organization Name and CAGE Code
- Facility Clearance Level (FCL)
- Facility Security Officer (FSO) name and contact information

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- Primary point of contact name, contact information, and clearance level
- b. Strategic Architecture Thrust Areas briefing (SECRET level)
 - Indicate capability to receive this over SIPR (Yes/No)
 - If not capable of receiving via SIPR, the briefing will be held for those respondents with the appropriate clearance level in conjunction with the 1:1 technical exchange
 - c. MDS-Next Characteristics briefing (CUI level)
 - Confirm primary point of contract name and email address to receive briefing via DoD SAFE
 - d. 1:1 technical exchange with subject matter experts from MDA (anticipated the week of 27 April 2026 – 15 May 2026)
 - Indicate preference for in-person or virtual exchange. In-person exchanges will be held in Huntsville, AL and are available at the unclassified level and higher. Virtual exchanges are limited to unclassified, CUI, and Secret (with SIPR capability) levels.
 - Indicate desired classification level
 - Identify all attendee names, contact information, and clearance level

Upon receipt, the Government will verify clearance information, coordinate access to the briefings, and schedule the 1:1 technical exchange.

10.0 WHITE PAPER SUBMISSION

White papers must be received via secured electronic submission by MDA NLT 1200 CST on **15 May 2026 at the following addresses:**

- Unclassified submissions: mda.mbx.sppa@mail.mil
- For submissions at the CUI and higher, please send an unclassified email to mda.mbx.sppa@mail.mil to coordinate submission instructions

White papers are limited to a maximum of 5 pages. Multiple submissions per respondent are permitted. There is no prescribed format. Clarity of thought and reasoning is valued over presentation style.